Gender processing in Basque-Spanish code-switching: evidence from eye-tracking

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Several studies on Spanish–English mixed Determiner Phrases (DPs) reported asymmetries between masculine and feminine gender in highly proficient Spanish-English bilinguals: masculine determiners appear with English nouns whose translation equivalent is both masculine or feminine (e.g., *el* toy, *el* cookie, Sp., *el juguete, la galleta*), whereas feminine determiners are only used with feminine translations (e.g., *la* cookie, Jake et al. 2002; Liceras et al. 2008). This evidence has led scholars to refer to the masculine gender as default (Beatty-Martínez & Dussias, 2019). Valdés Kroff et al. (2017) provided evidence for the real-time processing of mixed Spanish-English DPs. They found that the feminine determiner revealed facilitation effects, whereas masculine did not. Note that low-proficiency English learners do not show this anticipatory effect for either determiner (Dussias et al., 2013).

Research on Spanish–Basque DPs —Spanish gendered, Basque nongendered—reveals different gender assignment strategies. Parafita-Couto et al. (2015) found that feminine gender was preferred in the acceptability of Spanish-Basque mixed DPs. However, in a more recent study Munarriz-Ibarrola et al. (2022) showed no evidence for any default gender; instead, they observed two different strategies: L1 Spanish speakers favored determiners matching the translation equivalent (analogy) (e.g., *el*.DET.MASC *beso* 'arm.MASC'), whereas L1 Basque speakers relied on phonology, preferring feminine for –*a* endings (e.g., *la*.DET.FEM *eztia* 'honey.FEM') and masculine for –*o* endings (e.g., *el*.DET.MASC *hosto* 'leaf.FEM'). Such results might be influenced by the metalinguistic knowledge of the participants, thus motivating our current study with an online task.

This study extends the design of Valdés Kroff et al. (2017) with Spanish-Basque bilinguals by investigating the processing of Spanish–Basque mixed DPs through anticipatory effects (eye movements) and reaction times (keyboard responses). Fourteen Spanish-Basque highly proficient bilinguals participated (7 L1 Basque, 7 L1 Spanish; 11 female; mean age = 27, SD = 4.88) in an eye-tracking study employing the Visual World Paradigm. Participants completed 4 practice trials and 40 experimental trials (randomized). Participants were presented two images (target and distractor) on screen and after 1000 ms heard mixed DPs in a fixed carrier phrase (e.g., *Encuentra*.FIND *la*.DET.FEM *orratza* 'needle.FEM'). Distractors were phonological competitors in Basque (e.g., *orratza* 'needle.FEM' (target) vs *orrazia* 'comb.MASC' (distractor)), which share the first (two) syllable(s), but differ in the grammatical gender of their translation equivalents. There were three variables: gender of the Spanish determiner (el/la), matching gender of the translation equivalent (match/mismatch) and transparency of the noun's word-ending (transparent, *-a-o*/nontransparent, *e-i-u-consonants*).

Results revealed no significant differences in their RTs, (L1 Spanish speakers (mean = 3810ms) vs. L1 Basque speakers mean = 3948). However, L1 Spanish where faster than L1 Basque ($p = 0.0181$) in mixed DPs where gender matched the translation equivalent and the word-ending was transparent (e.g., *el*.MASC *aizto* 'knife.MASC'). Regarding eye-movements, in two-way-interactions, L1 Spanish exhibited anticipatory effects in gender and transparency ($p < .0001$) and in matching analogy ($p < 0.01$); however, L1 Basque speakers did not. These differences intensify in the three-way-interactions, where fixations displayed robust effects for gender × transparency × profile ($p < .0001$) and match × transparency × profile ($p < .0001$), confirming that the combination of morpho-phonological and grammatical cues substantially modulates target fixation in L1 Spanish. In contrast, L1 Basque show no sensitivity to any of those relationships (Figure 1). A time-course analysis (Figure 2) confirmed earlier target fixations in L1 Spanish speakers, as they have benefited more from gender, matching analogy and transparency cues (see Region 3500 ms), whereas L1 Basque initially inspected the distractor (Region 3500-4000 ms) and then fixated the target (Region 4250 ms), suggesting a competitor effect and inhibition effort; they also displayed an unexpected equal fixation pattern for target and distractor in an earlier time window (Region 2800-3200 ms). The contributions of this study for the literature on gender assignment strategies in mixed DPs will be further discussed.

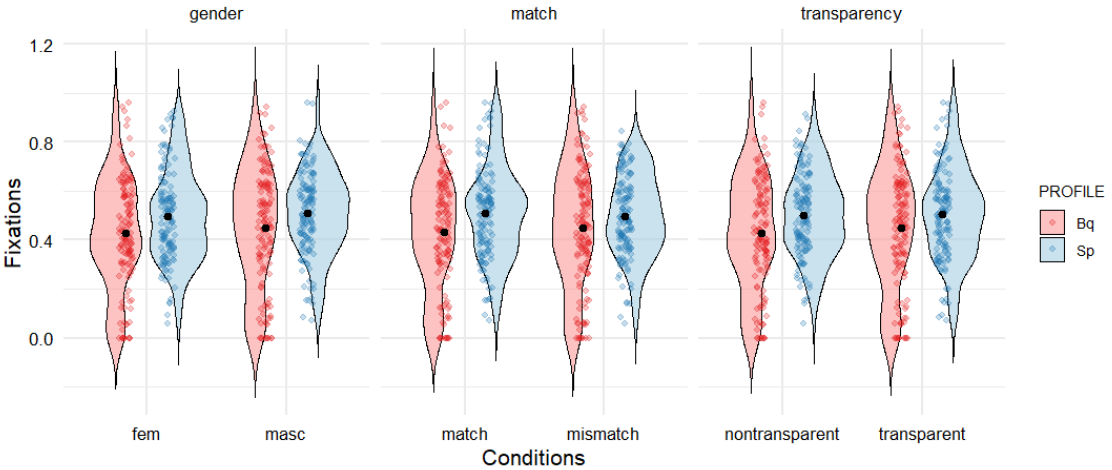

Figure 1. Fixations and mean fixations to target image by condition.

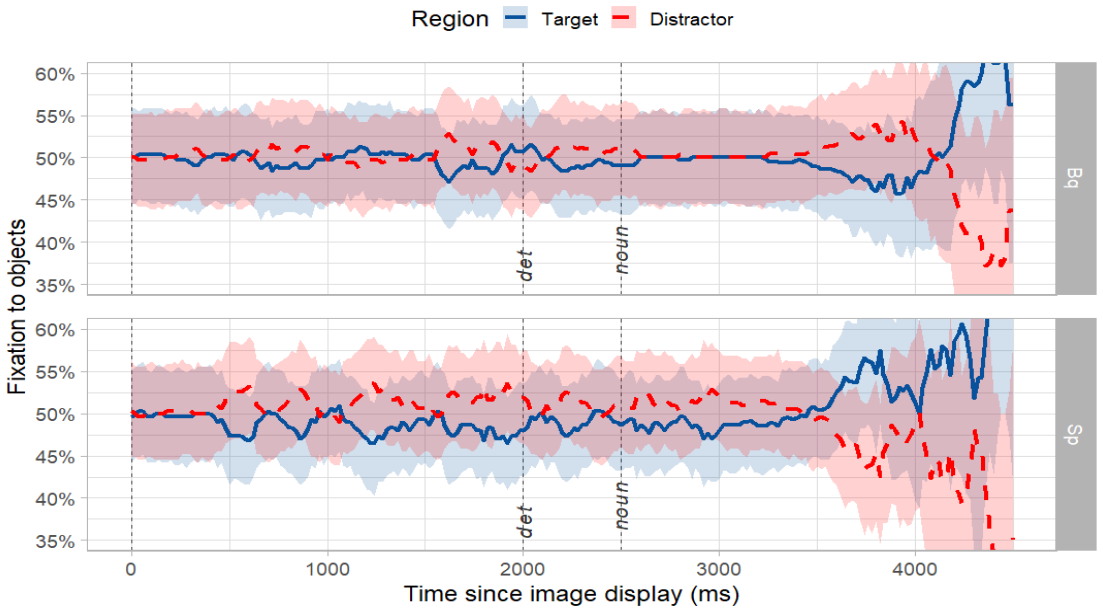

Figure 2. The proportion of total fixations to objects (targets and distractors). Determiner offset and noun offset are represented by a vertical dashed line.

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